



(12) **United States Patent**
Iwaki

(10) **Patent No.:** **US 12,414,280 B2**
(45) **Date of Patent:** **Sep. 9, 2025**

(54) **SUBSTRATE WORKING MACHINE**

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(*) Notice: Subject to any disclaimer, the term of this patent is extended or adjusted under 35 U.S.C. 154(b) by 413 days.

(21) Appl. No.: **17/906,308**

(22) PCT Filed: **Mar. 19, 2020**

(86) PCT No.: **PCT/JP2020/012266**

§ 371 (c)(1),

(2) Date: **Sep. 14, 2022**

(87) PCT Pub. No.: **WO2021/186666**

PCT Pub. Date: **Sep. 23, 2021**

(65) **Prior Publication Data**

US 2023/0093351 A1 Mar. 23, 2023

(51) **Int. Cl.**

H05K 13/08 (2006.01)

B25J 9/16 (2006.01)

B25J 13/08 (2006.01)

G06T 7/50 (2017.01)

G06T 7/70 (2017.01)

H05K 13/00 (2006.01)

H05K 13/04 (2006.01)

(52) **U.S. Cl.**

CPC **H05K 13/0812** (2018.08); **B25J 9/1687** (2013.01); **B25J 9/1697** (2013.01); **B25J 13/08** (2013.01); **G06T 7/50** (2017.01); **G06T 7/70** (2017.01); **H05K 13/0069** (2013.01); **G06T 2207/30141** (2013.01)

(58) **Field of Classification Search**

None

See application file for complete search history.

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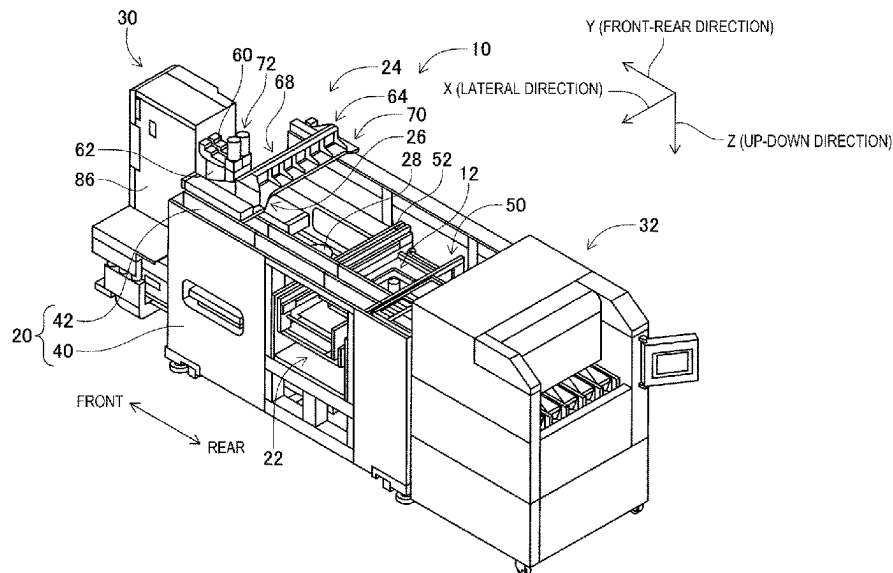
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(57) **ABSTRACT**

Provided is a substrate working machine including a holding device configured to hold a substrate in which multiple through-holes are formed, an inserting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the holding device, and an imaging device configured to simultaneously image a shape of a pair of pins and a pair of through-holes of the multiple through-holes, in which positions of the pair of through-holes and the shape of the pair of pins are calculated based on image data captured by the imaging device.

3 Claims, 14 Drawing Sheets



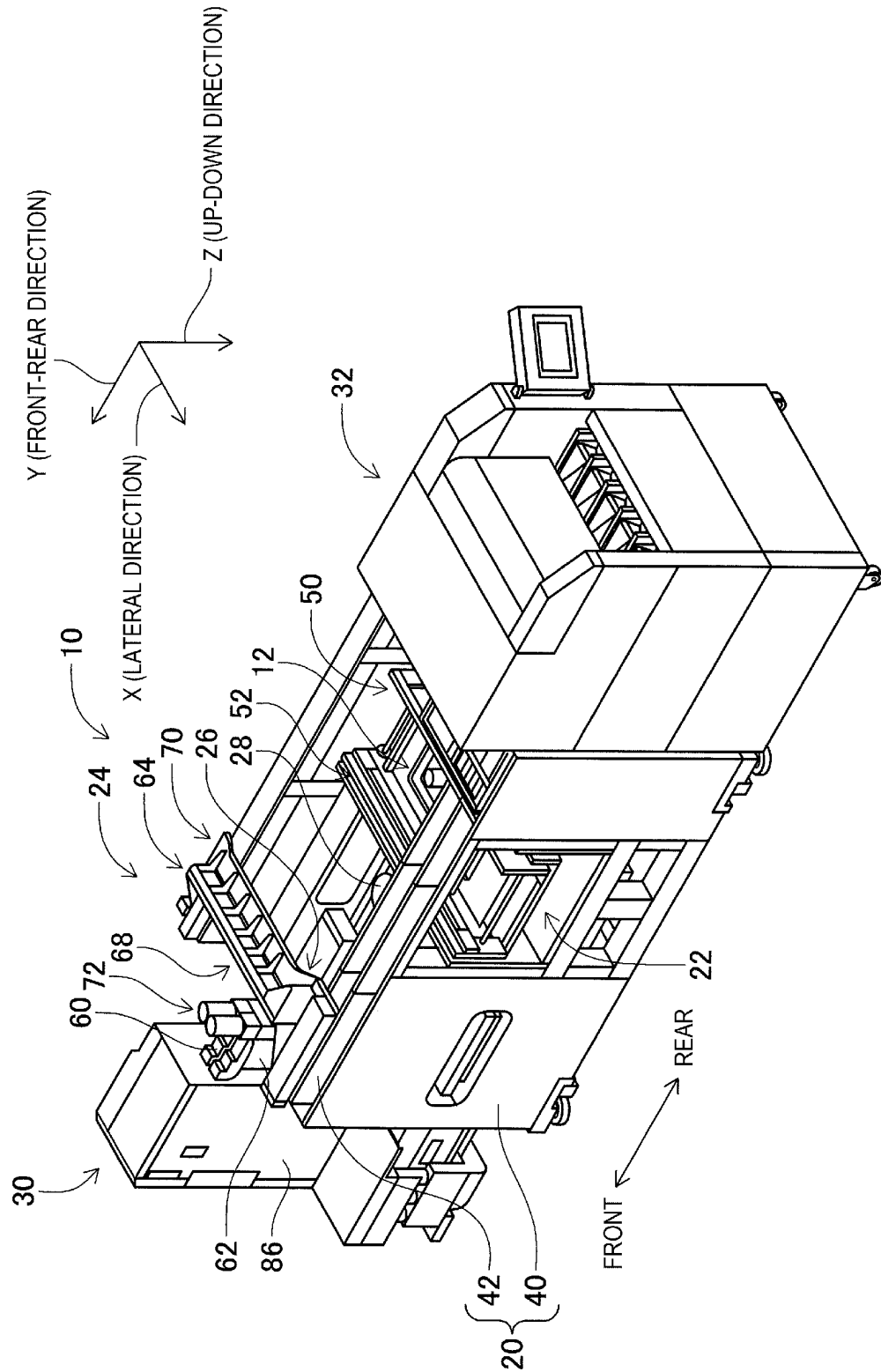


Fig. 1

Fig. 2

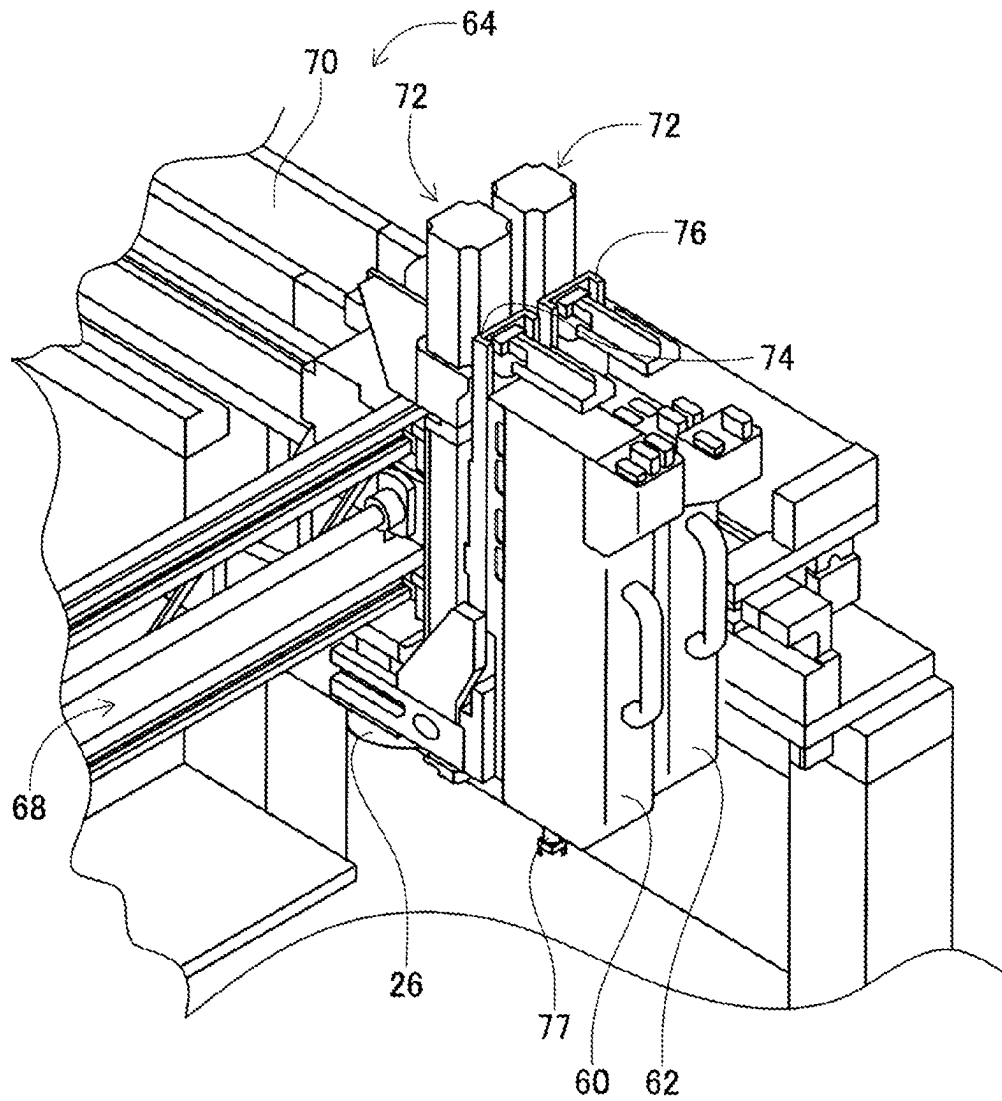


Fig. 3

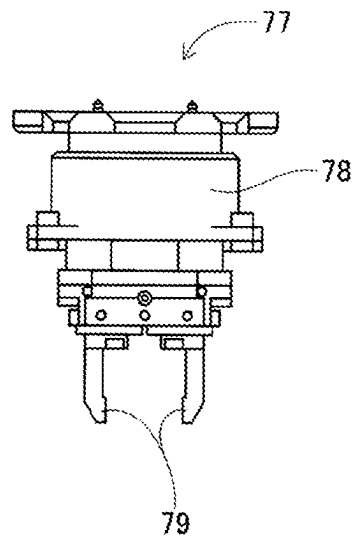


Fig. 4

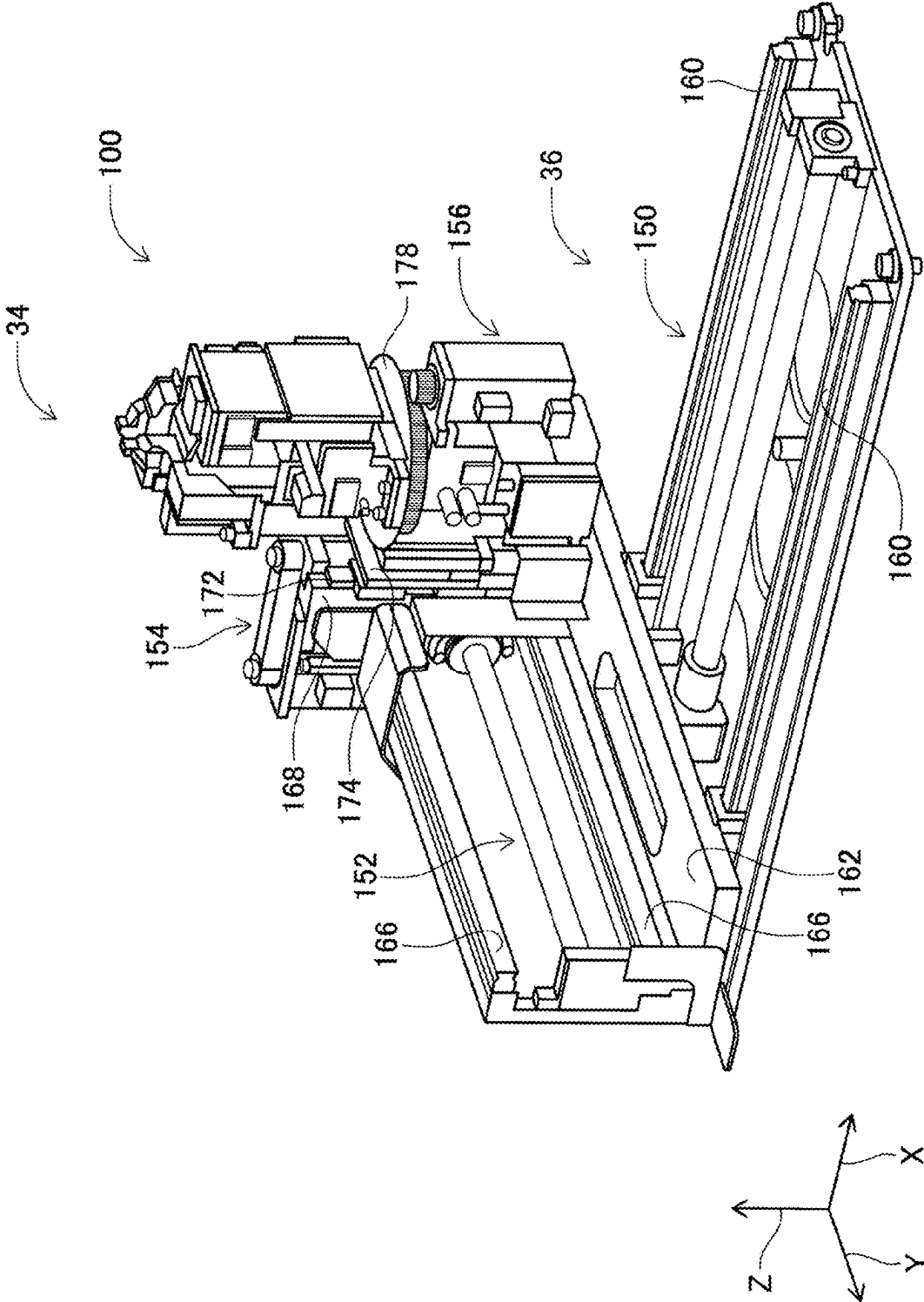


Fig. 5

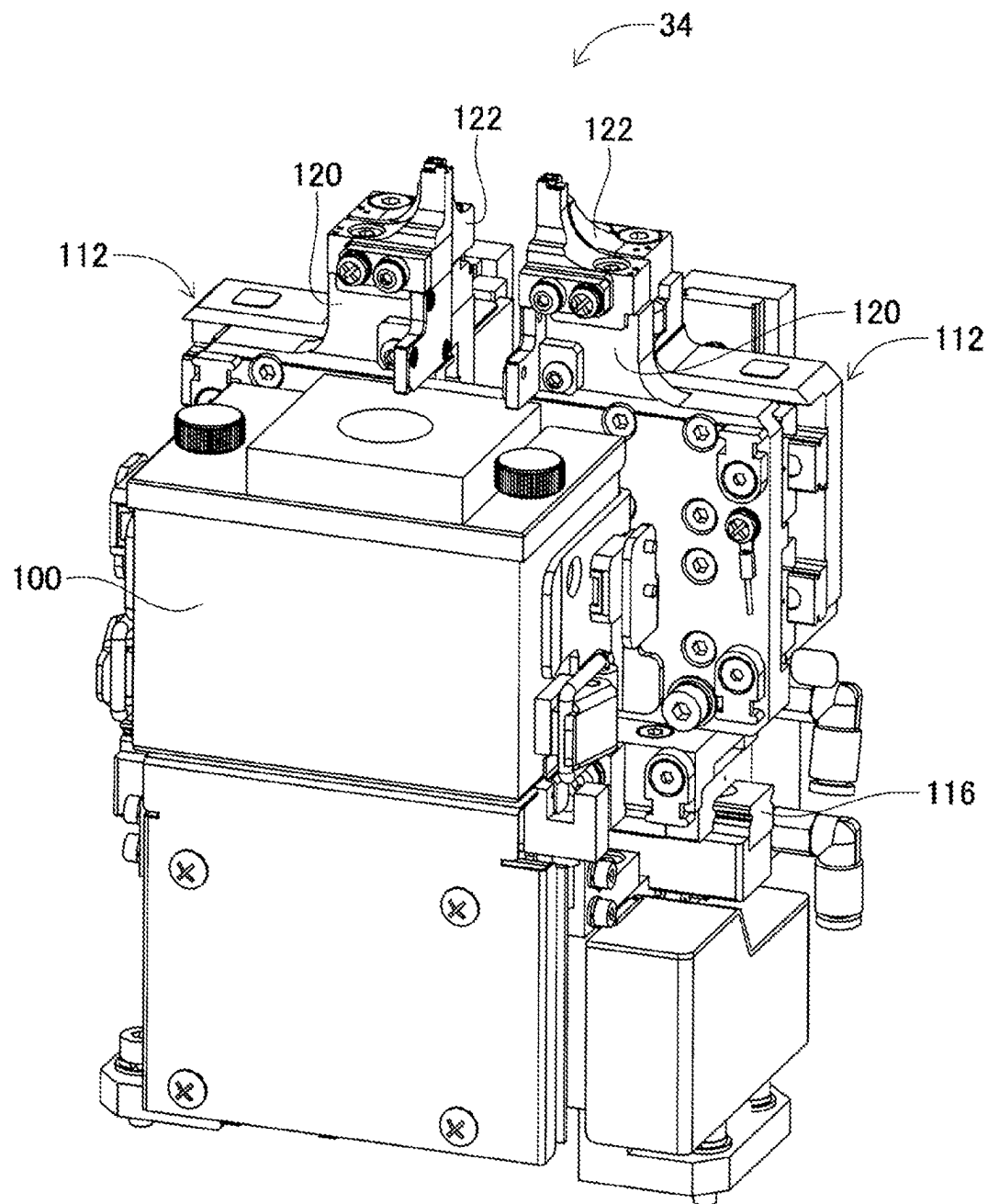


Fig. 6

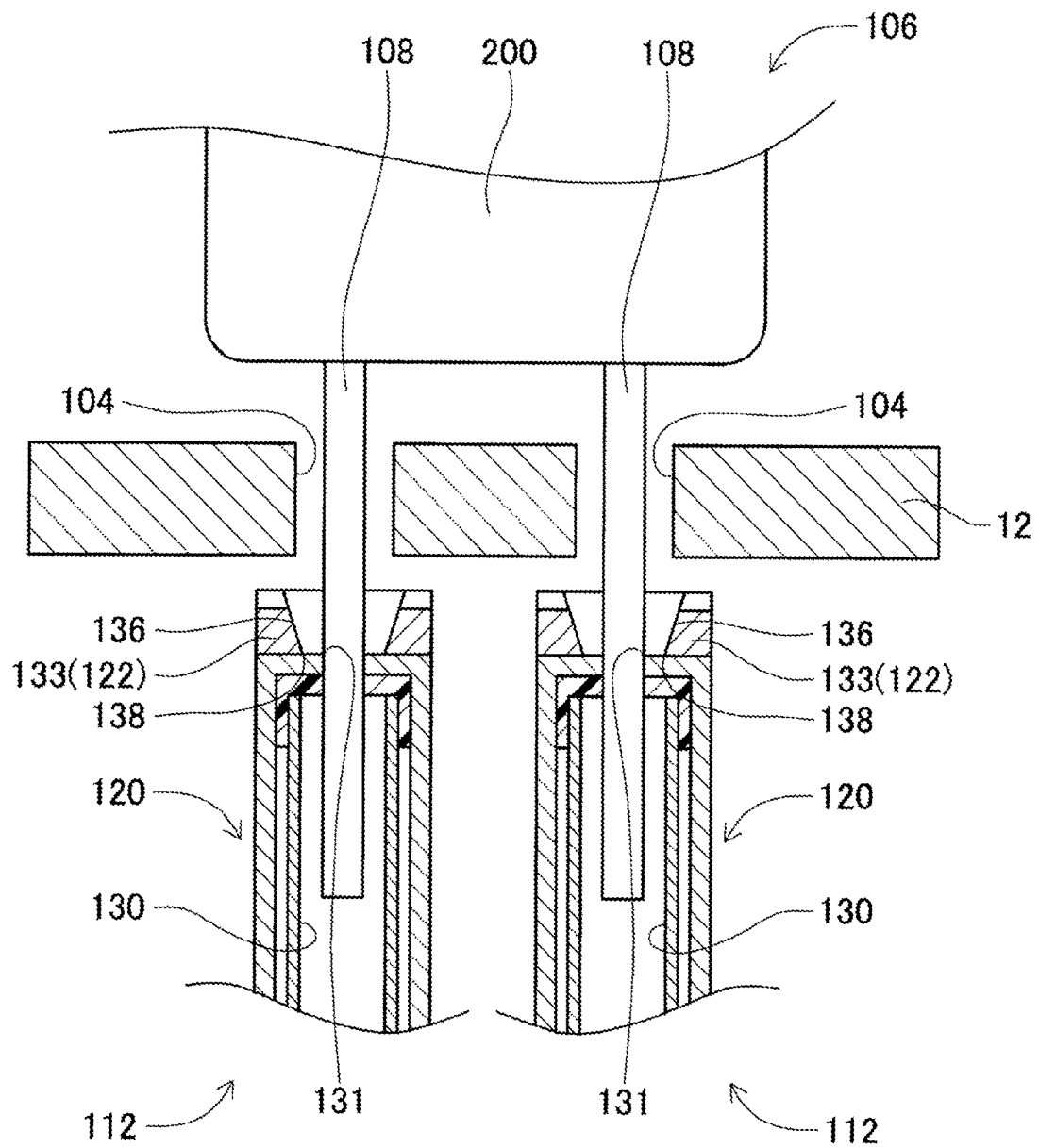


Fig. 7

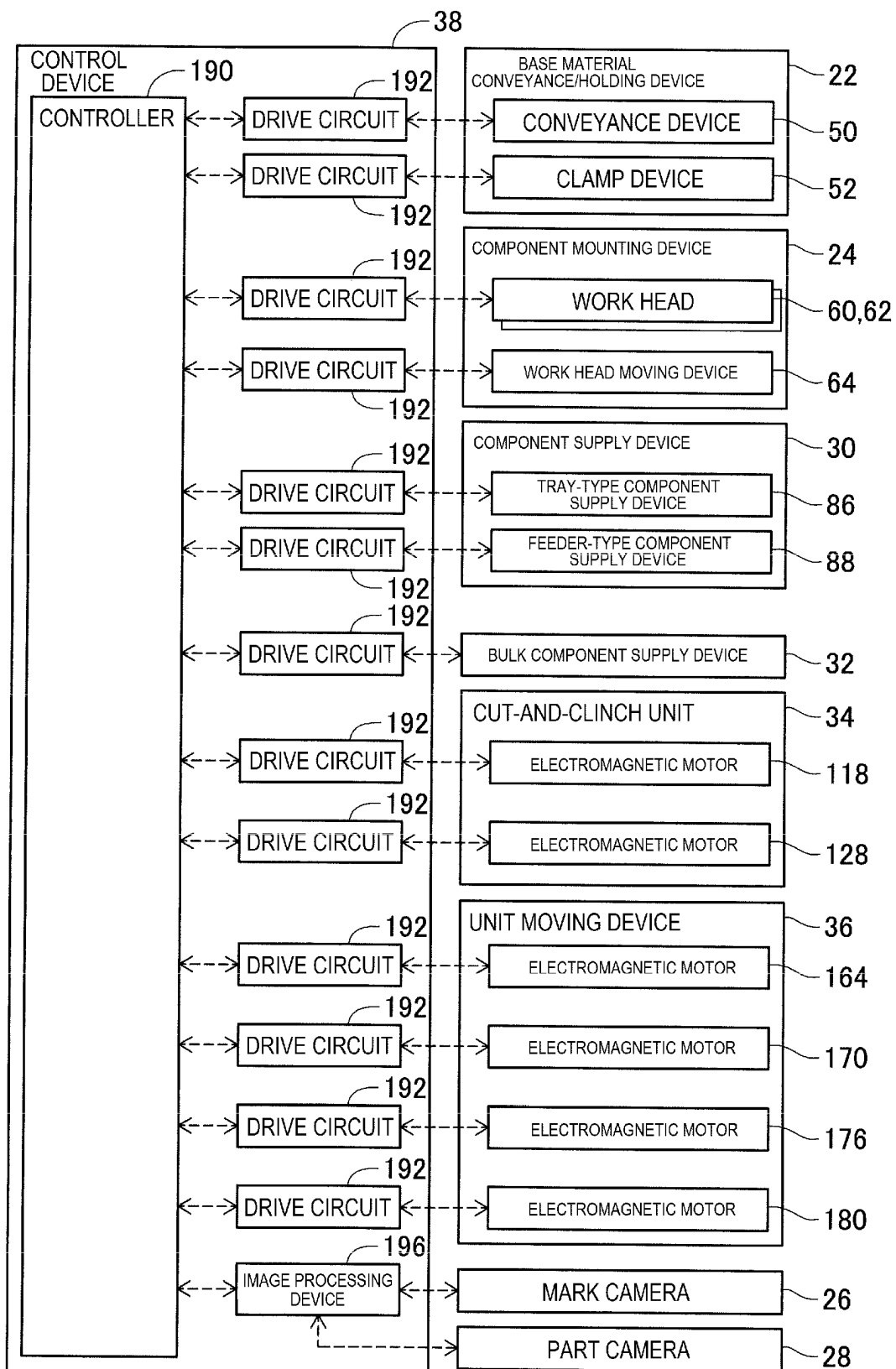


Fig. 8

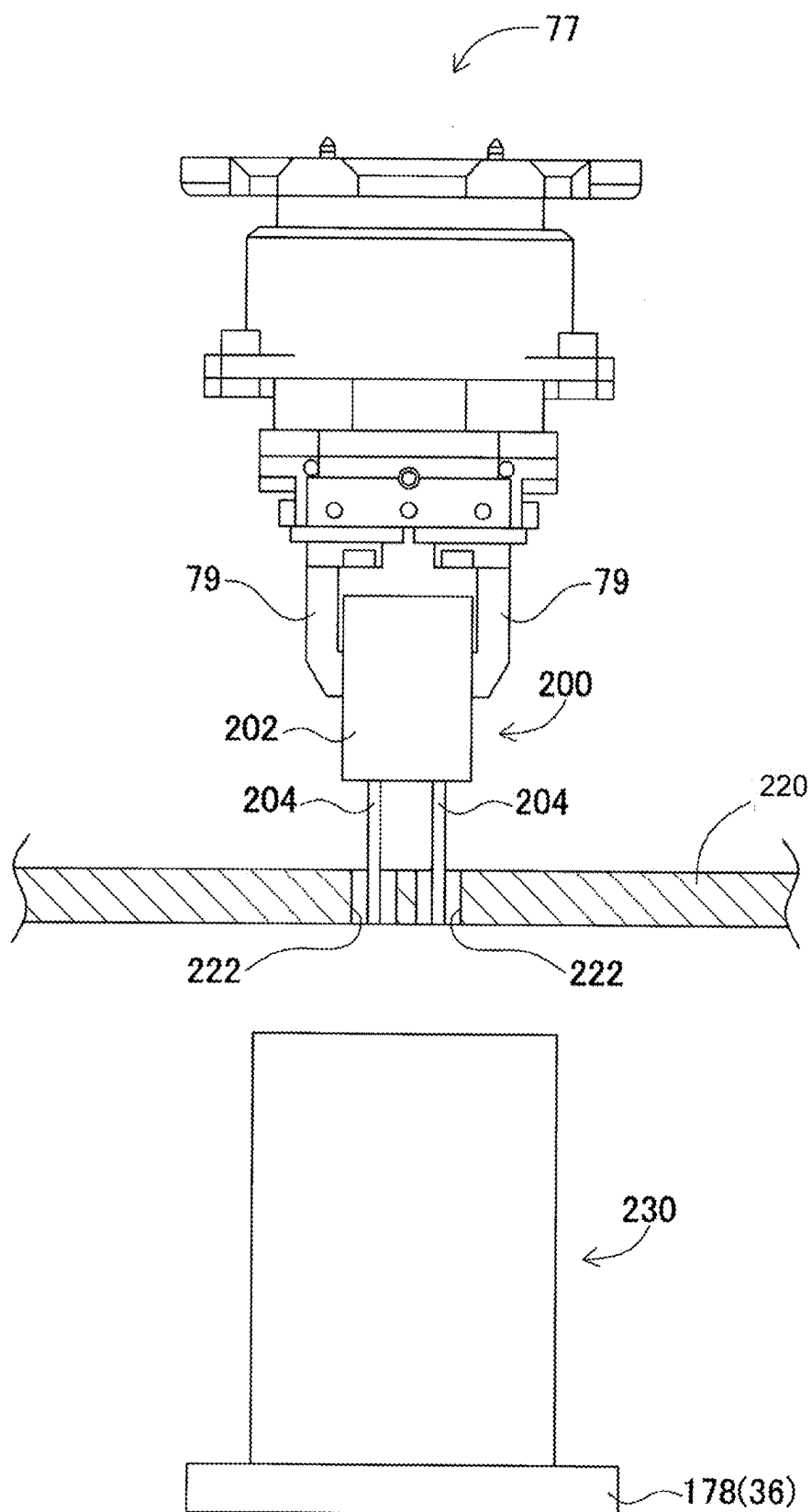


Fig. 9

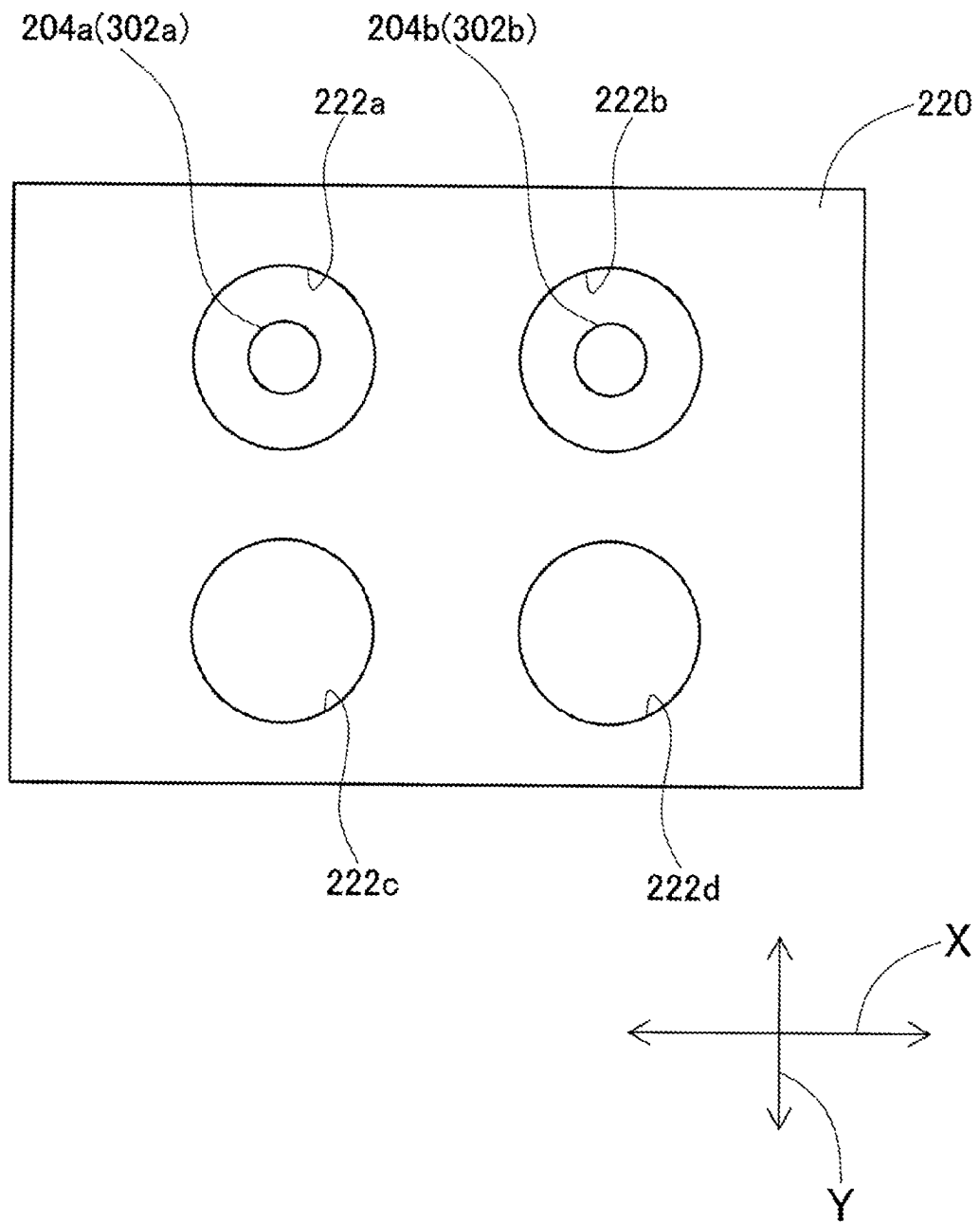


Fig. 10

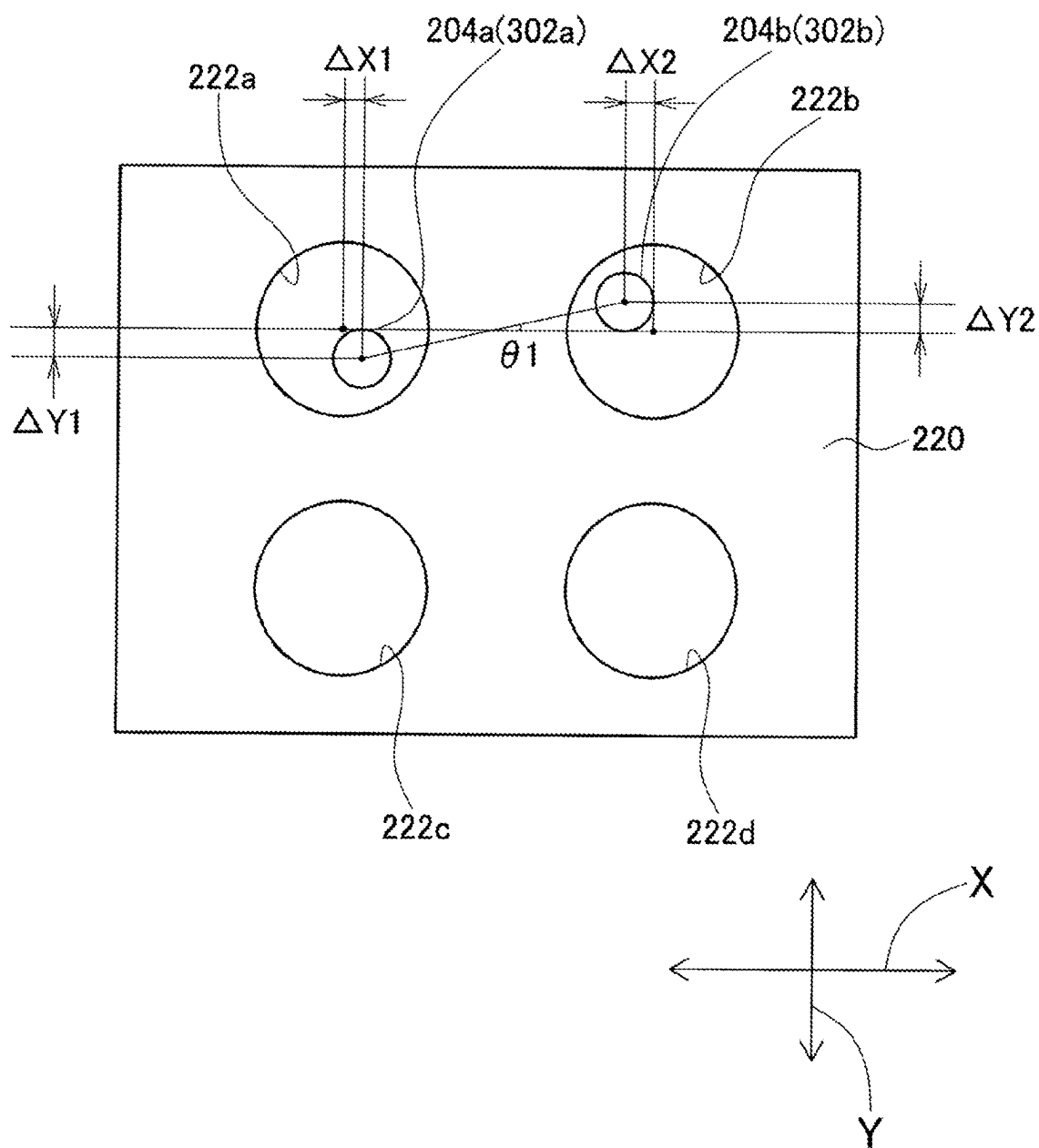


Fig. 11

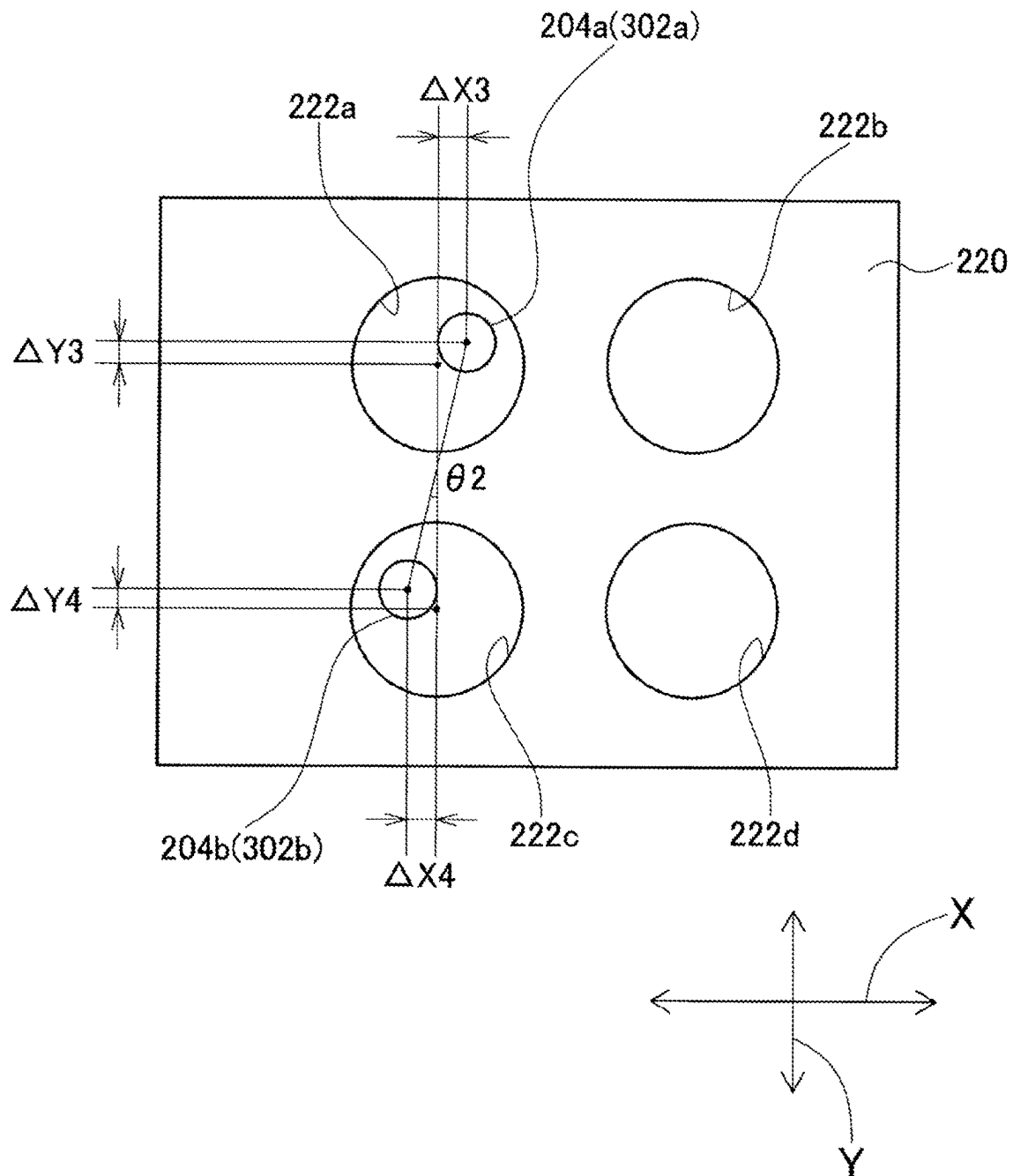


Fig. 12

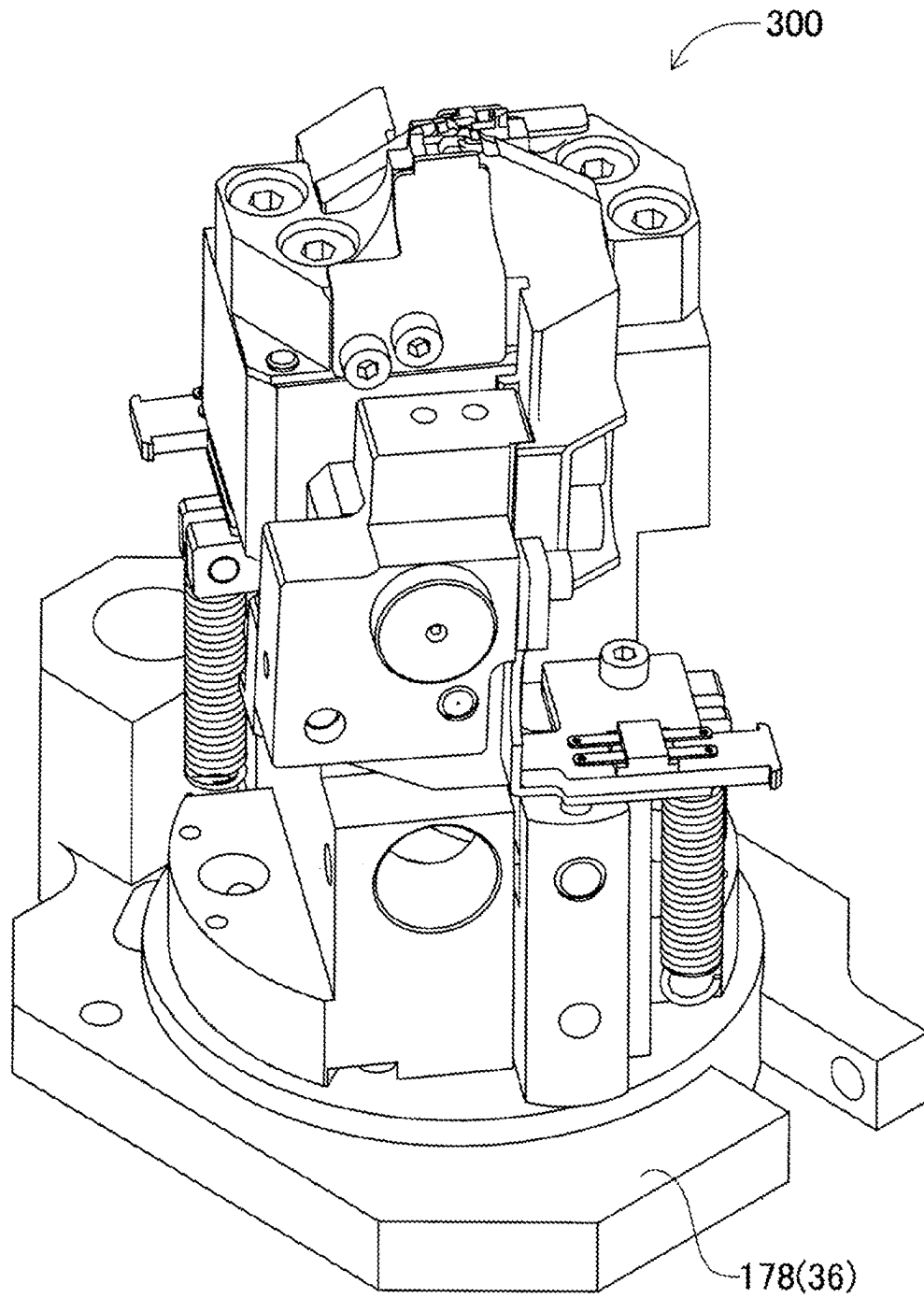


Fig. 13

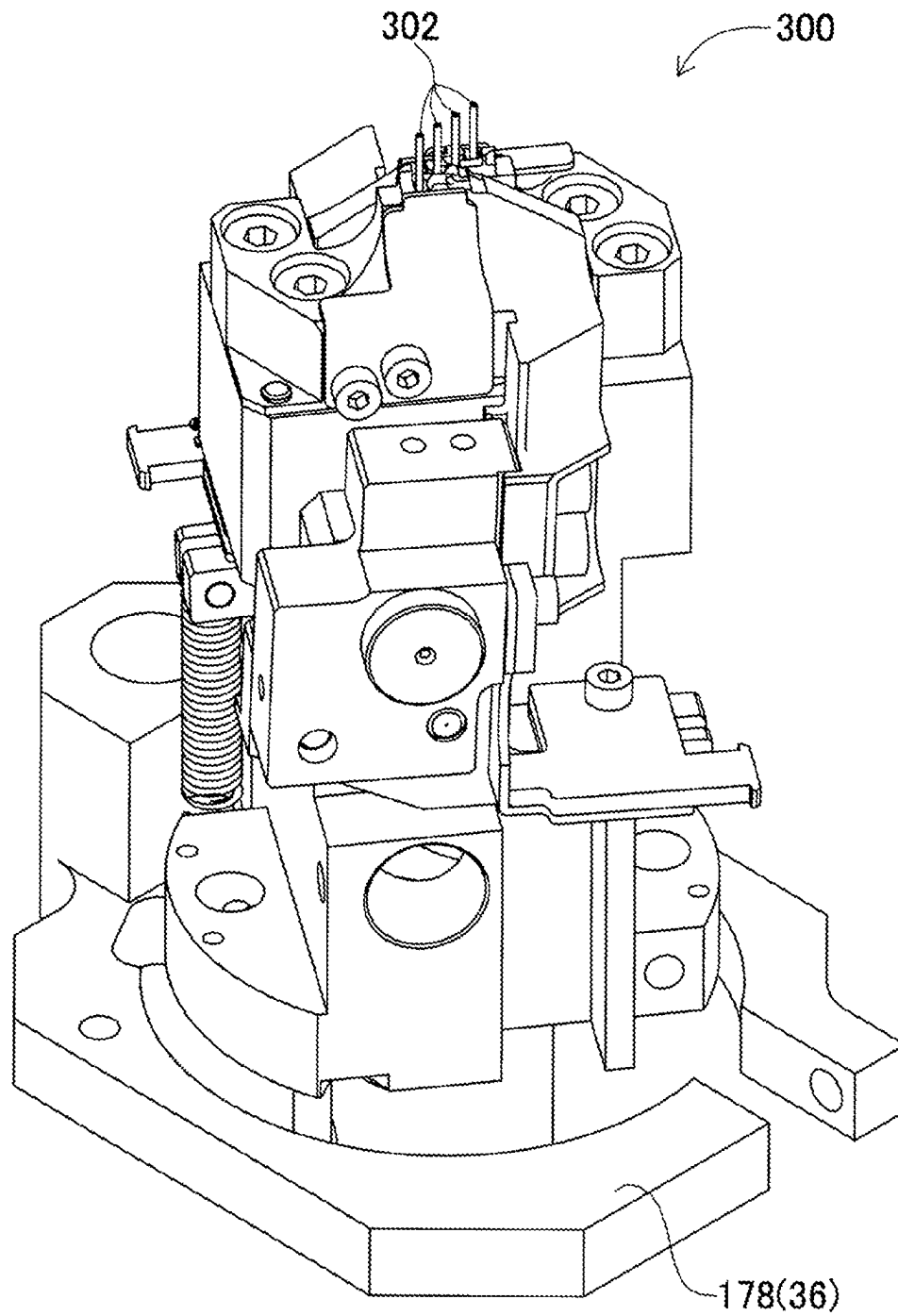


Fig. 14

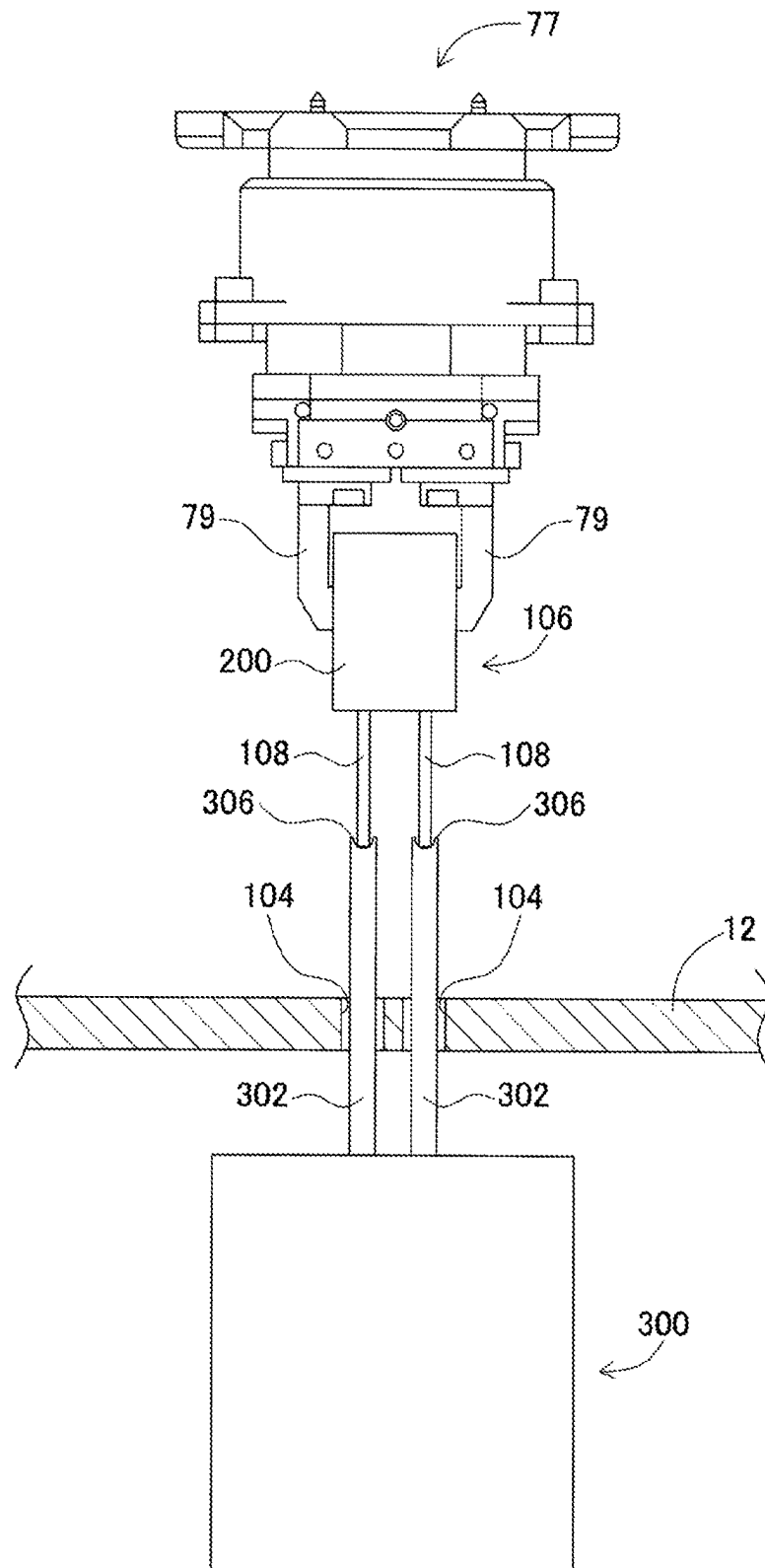
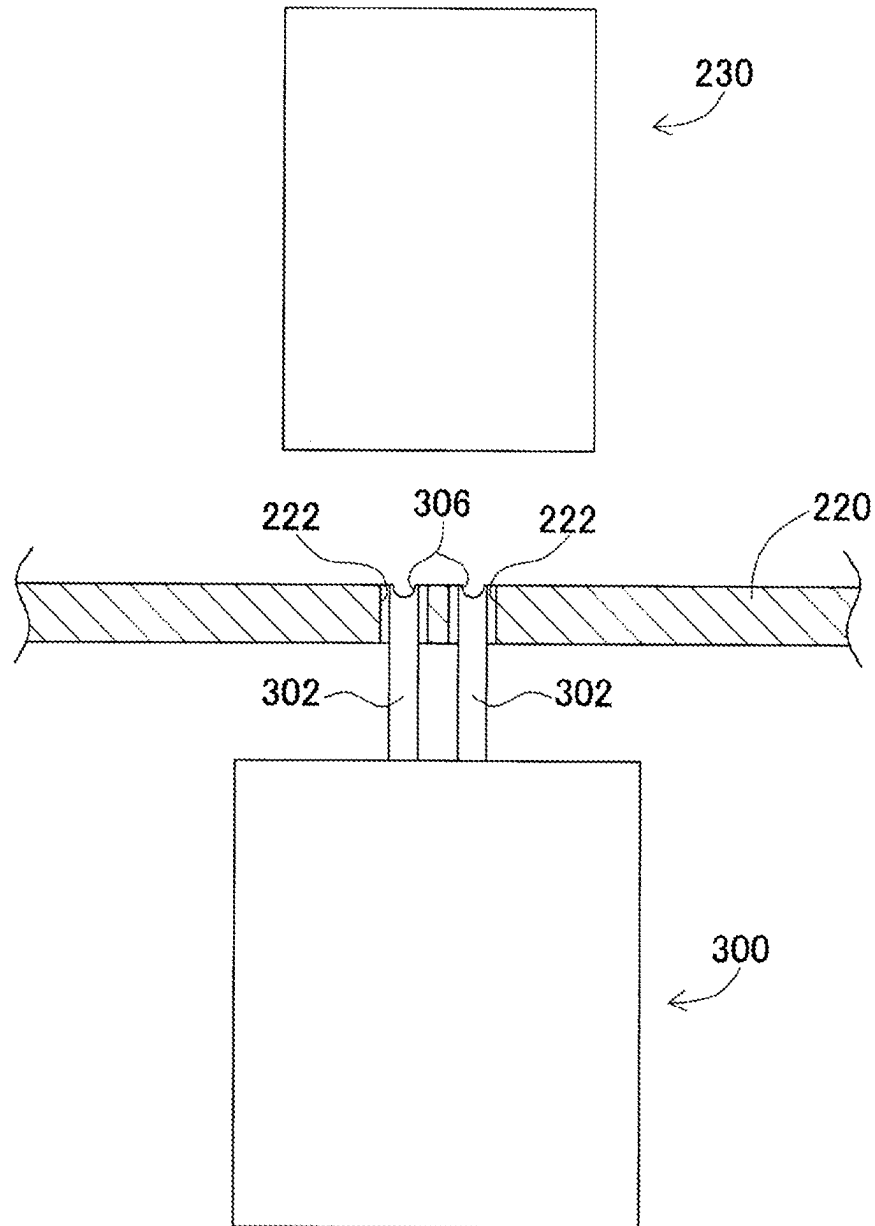


Fig. 15



1

SUBSTRATE WORKING MACHINE

TECHNICAL FIELD

The present disclosure relates to a substrate working machine for inserting multiple terminals of a component into multiple through-holes formed in a substrate.

BACKGROUND ART

In the following Patent Literature, a substrate working machine for inserting multiple terminals of a component into multiple through-holes formed in a substrate is described.

PATENT LITERATURE

Patent Literature 1: JP-A-H1-238197

BRIEF SUMMARY

Technical Problem

An object of the present specification is to insert multiple terminals of a component into multiple through-holes formed in a substrate.

Solution to Problem

In order to solve the above-mentioned problems, the present specification discloses a substrate working machine including a holding device configured to hold a substrate in which multiple through-holes are formed, an inserting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the holding device, and an imaging device configured to simultaneously image a shape of a pair of pins and a pair of through-holes of the multiple through-holes, in which positions of the pair of through-holes and the shape of the pair of pins are calculated based on image data captured by the imaging device.

Advantageous Effects

In the case of the substrate working machine of the present disclosure, the shape of the pair of pins and the pair of through-holes are simultaneously imaged and the positions of the pair of through-holes and the shape of the pair of pins are calculated based on the captured image data thereof. As a result, for example, it is possible to appropriately insert the multiple terminals into the multiple through-holes by calculating the positions of the pair of through-holes and the shape of the pair of pins based on the positions of the pair of through-holes and the shape of the pair of pins.

BRIEF DESCRIPTION OF DRAWINGS

FIG. 1 is a perspective view showing a component mounter.

FIG. 2 is a perspective view showing a component mounting device.

FIG. 3 is a view showing a component holding tool.

FIG. 4 is a perspective view showing a cut-and-clinch unit and a unit moving device.

FIG. 5 is a perspective view showing the cut-and-clinch unit.

2

FIG. 6 is a schematic view showing the cut-and-clinch unit during the cutting and bending of leads of a lead component.

FIG. 7 is a block diagram showing a control device.

FIG. 8 is a schematic view showing a state where a through-hole with pins inserted therein is imaged by an imaging device.

FIG. 9 is a captured image of a through-hole with pins inserted therein.

FIG. 10 is a captured image of a through-hole with pins inserted therein.

FIG. 11 is a captured image of a through-hole with pins inserted therein.

FIG. 12 is a perspective view showing a cut-and-clinch unit.

FIG. 13 is a perspective view showing the cut-and-clinch unit.

FIG. 14 is a schematic view showing a state where leads of a lead component are inserted into through-holes of a circuit base material by means of the cut-and-clinch unit.

FIG. 15 is a schematic view showing a state where a through-hole with guide pins inserted therein is imaged by the imaging device.

DESCRIPTION OF EMBODIMENTS

Hereinafter, exemplary embodiments of the present disclosure will be described in detail with reference to drawings.

FIG. 1 shows component mounter 10. Component mounter 10 is a device that performs a component mounting operation on circuit base material 12. Component mounter 10 includes device main body 20, base material conveyance and holding device 22, component mounting device 24, mark camera 26, part camera 28, component supply device 30, bulk component supply device 32, cut-and-clinch unit (refer to FIG. 5) 34, unit moving device (refer to FIG. 4) 36, and control device (refer to FIG. 7) 38. Examples of circuit base material 12 include a circuit substrate, a substrate having a three-dimensional structure, and the like, and examples of a circuit substrate include a printed wiring board, a printed circuit board, and the like.

Device main body 20 is composed of frame 40 and beam 42 suspended on frame 40. Base material conveyance and holding device 22 is disposed at the center of frame 40 in a front-rear direction, and has conveyance device 50 and clamp device 52. Conveyance device 50 is a device for conveying circuit base material 12, and clamp device 52 is a device for holding circuit base material 12. As a result, base material conveyance and holding device 22 conveys circuit base material 12 and fixedly holds circuit base material 12 at a predetermined position. In the description below, the conveyance direction of circuit base material 12 will be referred to as an X-direction, a horizontal direction perpendicular to the X-direction will be referred to as a Y-direction, and a vertical direction will be referred to as a Z-direction. That is, the width direction of component mounter 10 is the X-direction, and the front-rear direction is the Y-direction.

Component mounting device 24 is disposed on beam 42 and includes two work heads 60 and 62 and work head moving device 64. Work head moving device 64 includes X-direction moving device 68, Y-direction moving device 70, and Z-direction moving devices 72. In addition, two work heads 60 and 62 are integrally moved to any position on frame 40 by X-direction moving device 68 and Y-direction moving device 70. In addition, each of work heads 60

3

and 62 is positioned and mounted on sliders 74 and 76 such that each of work heads 60 and 62 can be detached and attached by a worker without using any tool, and Z-direction moving devices 72 individually move sliders 74 and 76 in an up-down direction. That is, work heads 60 and 62 are moved in the up-down direction individually by Z-direction moving devices 72.

In addition, as illustrated in FIG. 2, component holding tool 77 is attached to a lower end surface of each of work heads 60 and 62. Component holding tool 77 is a so-called chuck, and includes main body 78 and pair of claws 79 as illustrated in FIG. 3. Pair of claws 79 is disposed to extend downwards from a lower surface of main body 78 and linearly slides to move to and away from each other. As a result, component holding tool 77 holds a component with pair of claws 79 by causing pair of claws 79 to move to each other, and releases the component from between pair of claws 79 by causing pair of claws 79 to move away from each other. In addition, each of work heads 60 and 62 is provided with a rotation device (not illustrated) that rotates component holding tool 77 around the vertical axis, so that the posture of a component held by component holding tool 77 can be changed by means of the operation of the rotation device.

Mark camera 26 is attached to slider 74 in a state of facing downward on the vertical line as illustrated in FIG. 2, and mark camera 26 moves together with work head 60 in the X-direction, the Y-direction, and the Z-direction. As a result, mark camera 26 images any position on frame 40. As illustrated in FIG. 1, part camera 28 is disposed between base material conveyance and holding device 22 on frame 40 and component supply device 30 in a state of facing upward on the vertical line. As a result, part camera 28 images components held by component holding tools 77 of work heads 60 and 62.

Component supply device 30 is disposed at a first end portion of frame 40 in the front-rear direction. Component supply device 30 includes tray-type component supply device 86 and feeder-type component supply device (refer to FIG. 6) 88. Tray-type component supply device 86 is a device that supplies a component in a state in which the component is placed on a tray. Feeder-type component supply device 88 is a device that supplies a component by using a tape feeder or a stick feeder (not illustrated).

Bulk component supply device 32 is disposed at a second end portion of frame 40 in the front-rear direction. Bulk component supply device 32 is a device that aligns multiple components in a scattered state and supplies the components in an aligned state. That is, bulk component supply device 32 is a device that aligns multiple components in any posture to a predetermined posture, and supplies the components of the predetermined posture.

Examples of components which are supplied by component supply device 30 and bulk component supply device 32 include electronic circuit components, constituent components of a solar cell, constituent components of a power module, and the like. In addition, the electronic circuit components include a component with a lead, a component without a lead, and the like.

Cut-and-clinch unit 34 and unit moving device 36 are disposed below conveyance device 50. Cut-and-clinch unit 34 can be moved to any position below conveyance device 50 by unit moving device 36. Cut-and-clinch unit 34 is a device for cutting and bending leads (refer to FIG. 6) 108 of lead component (refer to FIG. 6) 106 that are inserted into through-holes 104 formed in circuit base material 12 (refer to FIG. 6). As illustrated in FIG. 5, cut-and-clinch unit 34

4

includes pair of sliding bodies 112. Pair of sliding bodies 112 are slidably supported by slide rail 116 disposed to extend in the X-direction. With this configuration, pair of sliding bodies 112 moves to and away from each other in the X-direction. In addition, the distance between pair of sliding bodies 112 is controllably changed by the driving of electromagnetic motor (refer to FIG. 7) 118.

In addition, each of pair of sliding bodies 112 includes fixed portion 120 and movable portion 122 and is slidably held by slide rail 116 at fixed portion 120. In addition, movable portion 122 is held by fixed portion 120 to be slidable in the X-direction. In addition, movable portion 122 is controllably slid in the X-direction with respect to fixed portion 120 by the driving of electromagnetic motor (refer to FIG. 7) 128.

In addition, as illustrated in FIG. 6, an upper end portion of fixed portion 120 has a tapered shape, and first insertion hole 130 is formed to penetrate the upper end portion in the up-down direction. In addition, an opening edge of first insertion hole 130 at an upper end surface is fixed blade 131. Meanwhile, an upper end portion of movable portion 122 also has a tapered shape, and bent portion 133, which is bent into an L-shape, is formed at the upper end portion. Bent portion 133 extends above an upper end surface of fixed portion 120. In addition, first insertion hole 130, which is open at the upper end surface of fixed portion 120, is covered by bent portion 133, and second insertion hole 136 is formed in bent portion 133 such that second insertion hole 136 faces first insertion hole 130. Note that an opening edge of second insertion hole 136 at a lower end surface of bent portion 133 is movable blade 138.

In addition, unit moving device 36, as illustrated in FIG. 4, includes X-direction moving device 150, Y-direction moving device 152, Z-direction moving device 154, and rotation device 156. X-direction moving device 150 includes slide rails 160 and X-slider 162. Slide rails 160 are disposed to extend in the X-direction, and X-slider 162 is slidably held by slide rails 160. In addition, X-slider 162 moves in the X-direction by the driving of electromagnetic motor (refer to FIG. 7) 164. Y-direction moving device 152 includes slide rails 166 and Y-slider 168. Slide rails 166 are disposed on X-slider 162 to extend in the Y-direction, and Y-slider 168 is slidably held by slide rails 166. In addition, Y-slider 168 moves in the Y-direction by the driving of electromagnetic motor (refer to FIG. 7) 170. Z-direction moving device 154 includes slide rails 172 and Z-slider 174. Slide rails 172 are disposed on Y-slider 168 to extend in the Z-direction, and Z-slider 174 is slidably held by slide rails 172. In addition, Z-slider 174 moves in the Z-direction by the driving of electromagnetic motor (refer to FIG. 7) 176.

In addition, rotation device 156 includes rotary table 178 having a substantially circular disc-like shape. Rotary table 178 is supported on Z-slider 174 such that rotary table 178 can rotate around a vertical axis thereof and rotates by the driving of electromagnetic motor (refer to FIG. 7) 180. In addition, on rotary table 178, cut-and-clinch unit 34 is positioned and disposed such that cut-and-clinch unit 34 can be detached and attached through a one-touch action performed by a worker without using any tool. With such a structure, cut-and-clinch unit 34 is moved to any position between a pair of conveyance lanes of base material conveyance and holding device 22, which is for conveyance of circuit base material 12, by X-direction moving device 150, Y-direction moving device 152, and Z-direction moving device 154 and is rotated to any angle by rotation device

5

156. As a result, cut-and-clinch unit 34 can be positioned at any position below circuit base material 12 held by clamp device 52.

As illustrated in FIG. 7, control device 38 includes controller 190, multiple drive circuits 192, and image processing device 196. Multiple drive circuits 192 are connected to conveyance device 50, clamp device 52, work heads 60 and 62, work head moving device 64, tray-type component supply device 86, feeder-type component supply device 88, bulk component supply device 32, and electromagnetic motors 118, 128, 164, 170, 176, and 180, which are described above. Controller 190 includes a CPU, a ROM, a RAM, and the like, is mainly composed of a computer, and is connected to multiple drive circuits 192. As a result, the operations of base material conveyance and holding device 22, component mounting device 24, and the like are controlled by controller 190. Furthermore, controller 190 is also connected to image processing device 196. Image processing device 196 is a device that processes image data obtained by mark camera 26 and part camera 28, and controller 190 acquires various pieces of information from the image data.

With the configuration as described above, component mounter 10 performs a component mounting operation on circuit base material 12 held by base material conveyance and holding device 22. Although component mounter 10 can mount various types of components on circuit base material 12, a case where lead component 106 is mounted on circuit base material 12 will be described below.

Specifically, circuit base material 12 is conveyed to a work position by conveyance device 50 of base material conveyance and holding device 22 and is fixedly held by clamp device 52 at the position. Next, mark camera 26 moves to a position above circuit base material 12 and images circuit base material 12. As a result, information on the holding position of circuit base material 12 and the like is obtained. In addition, component supply device 30 or bulk component supply device 32 supplies lead component 106 to work heads 60 and 62 at a predetermined supply position. Then, any of work heads 60 and 62 moves to a position above a component supply position and component main body (refer to FIG. 6) 200 of lead component 106 is gripped by pair of claws 79 of component holding tool 77.

Subsequently, work heads 60 and 62 holding the lead component 106 move to positions above part camera 28, and part camera 28 images lead component 106 held by pair of claws 79 of component holding tool 77. As a result, information on the holding position of the component and the like is obtained. Subsequently, work heads 60 and 62 holding lead component 106 move to positions above circuit base material 12 and errors of the holding position of circuit base material 12, errors of the holding position of the component, and the like are corrected. Then, leads 108 of lead component 106 held by component holding tool 77 are inserted into through-holes 104 formed in circuit base material 12. At this time, cut-and-clinch unit 34 is moved to a position below circuit base material 12.

Specifically, in cut-and-clinch unit 34, the distance between pair of sliding bodies 112 is moved and positioned by the operation of electromagnetic motor 118 such that the distance between second insertion holes 136 of pair of sliding bodies 112 is the same as the distance between two through-holes 104 formed in circuit base material 12. In addition, rotation device 156 is operated and positioned such that the direction of alignment of two through-holes 104 of circuit base material 12 coincides with the direction of alignment of two second insertion holes 136 of pair of sliding bodies 112.

6

Then, X-direction moving device 150 and Y-direction moving device 152 are operated and cut-and-clinch unit 34 is moved and positioned such that the XY-direction coordinates of second insertion holes 136 coincide with the XY-direction coordinates of through-holes 104 of circuit base material 12. As a result, cut-and-clinch unit 34 is moved along the XY-directions and second insertion holes 136 of sliding bodies 112 and through-holes 104 of circuit base material 12 are positioned in a state of overlapping with each other in the up-down direction.

Next, cut-and-clinch unit 34 is raised and positioned by the operation of Z-direction moving device 154 such that an upper surface of movable portion 122 comes into contact with a lower surface of circuit base material 12 or is positioned slightly lower than the lower surface of circuit base material 12. As described above, the operations of X-direction moving device 150, Y-direction moving device 152, Z-direction moving device 154, and rotation device 156 are controlled such that cut-and-clinch unit 34 is positioned below circuit base material 12 in a state where second insertion holes 136 of sliding bodies 112 and through-holes 104 of circuit base material 12 overlap with each other.

In addition, in a case where leads 108 of lead component 106 held by component holding tool 77 are inserted into through-holes 104 of circuit base material 12, distal end portions of leads 108 are inserted into first insertion holes 130 of fixed portions 120 via second insertion holes 136 of movable portions 122 of cut-and-clinch unit 34, as illustrated in FIG. 6. Next, in a case where the distal end portions of leads 108 are inserted into first insertion holes 130 of fixed portions 120, movable portions 122 of pair of sliding bodies 112 slide away from each other by the operation of electromagnetic motor 128. As a result, leads 108 are cut by fixed blades 131 of first insertion holes 130 and movable blades 138 of second insertion holes 136.

In addition, pair of movable portions 122 further slides away from each other even after leads 108 are cut. Therefore, new distal end portions of leads 108 due to the cutting are bent as movable portions 122 slide. As a result, pair of leads 108 is bent away from each other, so that lead component 106 is mounted on circuit base material 12 in a state where leads 108 are prevented from coming out of through-holes 104.

As described above, in component mounter 10, leads 108 of lead component 106 are inserted into through-holes 104 of circuit base material 12, and distal ends of leads 108 are cut and bent by cut-and-clinch unit 34. Therefore, although it is desired to suitably insert leads 108 into through-holes 104 of circuit base material 12, a method of improving the insertion accuracy has not been established in a conventional component mounter. Therefore, in the case of component mounter 10, a jig provided on which pins are formed is used and the pins of the jig held by work heads 60 and 62 are inserted into the through-holes of circuit base material 12 so that the accuracy of insertion of the leads of the lead component into the through-holes of the circuit base material is improved based on imaging data of the through-holes obtained in a state where the pins are inserted.

Specifically, as illustrated in FIG. 8, jig 200 is composed of component main body 202 and pair of pins 204. Component main body 202 is substantially block-shaped, and pair of pins 204 extends linearly and vertically from the bottom surface of component main body 202 in the same direction. Pair of pins 204 is standardized, and the pitch, the outer diameter, the length dimension, and the like of the pins 204 are set in advance. In other words, standardized pair of pins 204 extends in the vertical direction from a surface of

7

component main body 202 with high accuracy and the accuracy is secured. In addition, component main body 202 of jig 200 is clamped by pair of claws 79 of component holding tool 77, so that jig 200 is held by component holding tool 77.

In addition, substrate 220 serving as a jig used for jig 200 is also prepared. As illustrated in FIG. 9, four through-holes 222 are formed in substrate 220, and four through-holes 222 are arranged in two rows and two columns. Four through-holes 222 are also standardized, and the pitch of two of four through-holes 222 that are adjacent to each other, the inner diameters of the through-holes 222, and the like are accurately processed to be dimensions set in advance. Note that the pitch of two of four through-holes 222 that are adjacent to each other and the pitch of pair of pins 204 of jig 200 are processed to be the same as each other. Incidentally, the pitch of two through-holes 222 is the distance between the center of first through-hole 222 and the center of second through-hole 222, and the pitch of pair of pins 204 is the distance between the center of an end surface of first pin 204 and the center of an end surface of second pin 204. In addition, the inner diameter of through-hole 222 is approximately three times the outer diameter of pin 204.

Substrate 220 having such a structure is conveyed to a work position by conveyance device 50 and is clamped at the work position by clamp device 52. Then, substrate 220 positioned and fixed is imaged by mark camera 26, and the position of each through-hole 222 is calculated based on the imaging data thereof. In addition, jig 200 held by component holding tool 77 is imaged by part camera 28, and the position of the pair of pins is calculated based on the imaging data thereof. Then, work heads 60 and 62 are moved by the operation of work head moving device 64 such that pair of pins 204 coincides with two of four through-holes 222 in the up-down direction, that is, the XY-coordinates of pair of pins 204 coincide with the XY-coordinates of two through-holes 222. Subsequently, work heads 60 and 62 are lowered by the operation of work head moving device 64, so that pair of pins 204 is inserted into two through-holes 222 as illustrated in FIG. 8. At this time, work heads 60 and 62 are lowered such that end surfaces of pins 204 and a lower surface of substrate 220 coincide with each other in the up-down direction, that is, the end surfaces of pins 204 and the lower surface of substrate 220 are flush with each other.

In addition, on a lower surface side of substrate 220, one imaging device 230 is attached to rotary table 178 of unit moving device 36 instead of cut-and-clinch unit 34 attached thereto. As described above, the cut-and-clinch unit 34 can be positioned, detached from, and attached to rotary table 178 through a one-touch action performed by a worker without using any tool and imaging device 230 is also attachably and detachably attached to rotary table 178, from which cut-and-clinch unit 34 has been removed, in a state of being positioned while facing upward on the vertical line. Then, imaging device 230 attached to rotary table 178 is moved to a position below through-holes 222 of substrate 220 by the operation of unit moving device 36.

Therefore, pair of pins 204 inserted into two through-holes 222 of substrate 220 is imaged by imaging device 230. That is, pair of pins 204 and two through-holes 222 with pair of pins 204 inserted therein are simultaneously imaged by imaging device 230. Note that imaging device 230 is a two-dimensional imaging device including an imaging element in which cells are two-dimensionally arranged and images a three-dimensional object to obtain a two-dimensional image. Therefore, the depth of field of imaging device 230 is relatively narrow. Therefore, as described above,

8

lower end surfaces of pins 204 inserted into through-holes 222 and the lower surface of substrate 220 are made flush with each other and are positioned within the depth of field of the imaging device, so that an appropriate captured image of the lower end surfaces of pins 204 and through-holes 222 is acquired.

In a case where standardized pins 204 are inserted into standardized through-holes 222 in this manner, an ideal insertion state is a state where pair of pins 204 is inserted into the centers of two through-holes 222. That is, as described above, in a case where two through-holes 222 with pair of pins 204 inserted therein are imaged by one imaging device 230, ideally, the centers of pair of pins 204 and the centers of two through-holes 222 coincide with each other without being offset from each other in the imaging data thereof, as illustrated in FIG. 9. However, as illustrated in FIG. 10, there is a case where the centers of pair of pins 204 and the centers of two through-holes 222 are offset from each other because of a slight difference in product accuracy between component mounters, a decrease in insertion accuracy with time caused by use of component mounter 10, or the like.

Therefore, for each component mounter, the control device calculates the position of a pair of through-holes and the shape of a pair of pins based on the position of the pair of through-holes and the shape of the pair of pins, acquires a correction value for appropriately inserting terminals of a component held by a work head into multiple through-holes of a circuit base material, and can appropriately insert the multiple terminals into the multiple through-holes based on the acquired correction value. Therefore, in component mounter 10, the amount of deviation between inserted pins 204 and through-holes 222 is calculated based on imaging data of two through-holes 222 with pair of pins 204 inserted therein, and a correction value at the time of inserting the pair of pins into the pair of through-holes is acquired based on the calculated amount of deviation. That is, for example, in a case where the centers of pair of pins 204a and 204b and the centers of two through-holes 222a and 222b are offset from each other as illustrated in FIG. 10, the amount of deviation ($\Delta X1$, $\Delta Y1$) in the XY-directions between the center of first pin 204a and the center of first through-hole 222a and the amount of deviation ($\Delta X2$, $\Delta Y2$) in the XY-directions between the center of second pin 204b and the center of second through-hole 222b are calculated. In addition, the angle of deviation (81) formed by a straight line connecting the center of first pin 204a and the center of second pin 204b and a straight line connecting the center of first through-hole 222a and the center of second through-hole 222b is also calculated.

Subsequently, through-holes 222 into which pair of pins 204 is inserted are changed and other pair of through-holes 222 is imaged. For example, component holding tool 77 holding jig 200 is rotated by 90 degrees by the rotation device. Then, according to the above-described procedure, pair of pins 204 is inserted into two through-holes 222. In the case of component holding tool 77 rotated by 90 degrees by the rotation device, since pair of pins 204 of held jig 200 also rotates by 90 degrees, as illustrated in FIG. 11, first pin 204a is inserted into first through-hole 222a and second pin 204b is inserted into second through-hole 222c. Then, pair of pins 204a and 204b and two through-holes 222a and 222c with the pair of pins inserted therein are simultaneously imaged by one imaging device 230, and the amount of deviation between pins 204 and through-holes 222 is calculated. That is, the amount of deviation ($\Delta X3$, $\Delta Y3$) in the XY-directions between the center of first pin 204a and the center of first

through-hole **222a** and the amount of deviation ($\Delta X4$, $\Delta Y4$) in the XY-directions between the center of second pin **204b** and the center of second through-hole **222c** are calculated. In addition, the angle of deviation ($\theta 2$) formed by a straight line connecting the center of pair of pins **204a** and the center of pin **204b** and a straight line connecting the center of pair of through-holes **222a** and the center of through-hole **222c** is also calculated.

Furthermore, through-holes **222** into which pair of pins **204** is inserted are changed and other pair of through-holes **222** is imaged. For example, component holding tool **77** holding jig **200** is further rotated by 90 degrees by a rotation device. Then, according to the above-described procedure, pair of pins **204** is inserted into two through-holes **222**. In the case of component holding tool **77** further rotated by 90 degrees by the rotation device, since pair of pins **204** of held jig **200** also rotates by 90 degrees, although not shown, first pin **204a** is inserted into first through-hole **222b** and second pin **204b** is inserted into second through-hole **222a**. Then, pair of pins **204a** and **204b** and two through-holes **222a** and **222c** with the pair of pins inserted therein are simultaneously imaged from below by one imaging device **230**, and the amount of deviation between pins **204** and through-holes **222** is calculated. That is, the amount of deviation in the XY-directions between the center of pin **204a** and the center of through-hole **222b** and the amount of deviation in the XY-directions between the center of the pin **204b** and the center of through-hole **222a** are calculated. In addition, similarly, an angle formed by a straight line connecting the center of pair of pins **204a** and the center of pin **204b** and a straight line connecting the center of pair of through-holes **222a** and the center of through-hole **222b** is also calculated.

Similarly, the amount of deviation and the angle of deviation between pins **204** inserted and the through-holes are calculated, and a correction value at the time of inserting a pair of pins into the pair of through-holes is acquired based on the calculated amount of deviation and angle of deviation. By using this correction value, it is possible to insert pair of pins **204a** and **204b** into the centers of two through-holes **222a** and **222b** as illustrated in FIG. 9, in a case where pair of pins **204a** and **204b** is to be inserted into two through-holes **222a** and **222b**, for example. In other words, pair of pins **204** can be inserted into two through-holes **222** in a state where the amount of deviation of the pins is zero. As a result, for an actual insertion operation, that is, the component mounter, it is possible to improve the insertion accuracy at the time of inserting pair of leads **108** of lead component **106**, of which the component main body has been held by a work head, into pair of through-holes **104** of circuit base material **12** positioned at the work position of the substrate conveyance device.

Note that the imaging of pair of through-holes **222** with pair of pins **204** inserted therein, calculation of the amount of deviation and the angle of deviation based on imaging data acquired by the imaging of pair of through-holes **222**, acquisition of a correction value for insertion of the pair of pins into the pair of through-holes that is acquired based on the calculated amount of deviation and angle of deviation, or the like is performed at the time of inspection of the accuracy of component mounter **10** or at the time of maintenance, inspection, or the like of component mounter **10** after delivery to a user. As a result, it is possible to stabilize the quality of component mounter **10**, avoid problems, maintain the insertion accuracy by periodic maintenance, and the like.

In addition, in component mounter **10**, instead of cut-and-clinch unit **34**, it is also possible to attachably and detach-

ably position and attach, to rotary table **178** of unit moving device **36**, a lead receiving type cut-and-clinch unit receiving a lead component inserted into an insertion hole of a circuit base material. Lead receiving type cut-and-clinch unit **300** is an existing device and when briefly described, as illustrated in FIGS. **12** and **13**, cut-and-clinch unit **300** includes four guide pins **302** that controllably extend from an upper end portion of cut-and-clinch unit **300**.

Four guide pins **302** are disposed to extend in the vertical direction and are arranged at equal pitches on one straight line. Each of four guide pins **302** controllably expands and contracts in length, and is stored in the inside of cut-and-clinch unit **300** in the most contracted state. In addition, a cutting device (not illustrated) for cutting leads of a lead component and a bending device (not illustrated) for bending the leads are stored in the inside of cut-and-clinch unit **300**. With such a structure, in cut-and-clinch unit **300** as well, it is possible to cut and bend leads of a lead component inserted into through-holes **104** of circuit base material **12**.

Specifically, the operation of rotation device **156** is controlled such that the direction of alignment of two through-holes **104** of circuit base material **12** coincides with the direction of alignment of four guide pins **302**. Then, by the operation of X-direction moving device **150** and Y-direction moving device **152**, cut-and-clinch unit **300** is moved below the circuit base material **12** and the positioning thereof is stopped such that the coordinates in the XY-directions of two through-holes **104** of circuit base material **12** coincide with the coordinates in the XY-directions of distal ends of two of four guide pins **302** that are to be inserted into two of the four through-holes. As a result, cut-and-clinch unit **300** is moved along the XY-directions and two through-holes **104** of circuit base material **12** and the distal ends of two of four guide pins **302** are positioned in a state of overlapping with each other in the up-down direction.

Next, cut-and-clinch unit **300** extends two through-holes **104** of the circuit base material and two of the four guide pins **302** that are in a state of overlapping with the two through-holes in the up-down direction. As a result, as illustrated in FIG. **14**, two guide pins **302** are inserted through two through-holes **104** in a direction from a lower side to an upper side and upper ends of two guide pins **302** protrude from the upper surface of circuit base material **12**. In addition, recessed portion **306** is formed on an upper end surface of each guide pin, and the operation of work head moving device **64** is controlled such that a lower end of each of leads **108** of lead component **106** is inserted into recessed portion **306**. That is, the operations of X-direction moving device **68** and Y-direction moving device **70** are controlled such that the coordinates in the XY-directions of the lower ends of pair of leads **108** of lead component **106** held by component holding tool **77** coincide with the coordinates in the XY-directions of the recessed portions of two guide pins **302** exposed at the upper surface of the circuit base material **12**. Then, the operation of Z-direction moving device **72** is controlled such that work heads **60** and **62** are lowered and the lower ends of pair of leads **108** of lead component **106** held by component holding tool **77** are inserted into recessed portions **306** of two guide pins **302** protruding from the upper surface of the circuit base material **12**.

Subsequently, cut-and-clinch unit **300** contracts two guide pins **302** protruding and extending from the upper surface of the circuit base material **12**. At this time, work heads **60** and **62** are also lowered by the operation of Z-direction moving device **72** at the same speed as the speed of contraction of guide pins **302**. As a result, two guide pins **302** contract with the lower ends of pair of leads **108** inserted into recessed

11

portions 306 of two guide pins 302. Then, as guide pins 302 contract, upper end surfaces of guide pins 302, that is, recessed portions 306 pass through through-holes 104 of circuit base material 12 and are stored in the inside of cut-and-clinch unit 300. At this time, the lower ends of pair of leads 108 inserted into recessed portions 306 also pass through pair of through-holes 104 of circuit base material 12 and are stored in the inside of cut-and-clinch unit 300. That is, with the lower ends of pair of leads 108 of lead component 106 guided by pair of guide pins 302, the lower ends of pair of leads 108 are inserted into pair of through-holes 104 of circuit base material 12 and are stored in the inside of cut-and-clinch unit 300. Then, the lower ends of pair of leads 108 stored in the inside of cut-and-clinch unit 300 are cut and bent by the cutting device. As a result, lead component 106 is mounted on circuit base material 12 in a state where leads 108 are prevented from coming out of through-holes 104.

As described above, in the case of cut-and-clinch unit 300, guide pins 302 are extended to be inserted into pair of through-holes 104 from a position below the circuit base material 12 and the distal ends of pair of guide pins 302 protrude from the upper surface of circuit base material 12. Then, guide pins 302 contract with the lower ends of pair of leads 108 of lead component 106 inserted into recessed portions 306 of the distal ends of guide pins 302, the lower ends of pair of leads 108 are stored in the inside of cut-and-clinch unit 300, and the pair of leads are cut and bent. Therefore, although it is desired to suitably insert guide pins 302 into through-holes 104 of circuit base material 12, a method of improving the accuracy of insertion of guide pins 302 has not been established in a conventional component mounter. Therefore, in the case of component mounter 10, similarly to the method of improving the accuracy of insertion of leads 108 into through-holes 104 described above, the accuracy of insertion of guide pins 302 is improved by means of imaging data.

Specifically, first, one imaging device 230 is attached to any of sliders 74 and 76 from which a work head has been removed by means of a mechanism for mounting work heads 60 and 62 instead of any of work heads 60 and 62. As described above, work heads 60 and 62 are positioned and mounted to sliders 74 and 76 respectively such that work heads 60 and 62 can be detached and attached through a one-touch action, and imaging device 230 is positioned and attached to any of sliders 74 and 76 from which work heads 60 and 62 have been removed in a state of facing downward on the vertical line. As described above, imaging device 230 can be attachably and detachably positioned and mounted to rotary table 178 as well and is positioned with high compatibility and attachably and detachably mounted to all of rotary table 178 and sliders 74 and 76.

Similarly to substrate 220 into which pair of pins 204 of jig 200 are inserted, a substrate into which pair of guide pins 302 of cut-and-clinch unit 300 is inserted is also prepared. Here, the description will be made on the assumption that substrate 220 is used not only for jig 200 but also for guide pins 302 of cut-and-clinch unit 300. That is, the pitch of two of four through-holes 222 that are adjacent to each other and the pitch of two of four guide pins 302 of the cut-and-clinch unit that are adjacent to each other are the same as each other, four through-holes 222 being formed in substrate 220. Incidentally, the pitch of two guide pins 302 is a distance between the center of a distal end of first guide pin 302 and the center of a distal end of second guide pin 302. In

12

addition, the inner diameter of one through-hole 222 is approximately three times the outer diameter of one guide pin 302.

Substrate 220 having such a structure is conveyed to a work position by conveyance device 50 and is clamped at the work position by clamp device 52. Then, by imaging device 230 attached to sliders 74 and 76, through-holes of substrate 220 that are adjacent to each other are imaged from above, and the coordinates in the XY-directions of two of four through-holes 222 that are adjacent to each other are calculated based on imaging data thereof. Meanwhile, in the case of cut-and-clinch unit 300 positioned below substrate 220, according to the procedure described above, the cut-and-clinch unit is moved and positioned by the operation of unit moving device 36 such that the distal ends of two of four guide pins 302 that are to be inserted into two of the four through-holes of which the coordinates have been calculated coincide with the calculated coordinates in the XY-directions of two through-holes 222 of the substrate that are adjacent to each other. Cut-and-clinch unit 300 extends two guide pins 302. As a result, two guide pins 302 are inserted into two through-holes 104, of which the coordinates coincide with the coordinates of two guide pins 302, from below. At this time, as illustrated in FIG. 15, cut-and-clinch unit 300 extends guide pins 302 such that the upper end surfaces of guide pins 302 coincide with an upper surface of the substrate 220 in the up-down direction, that is, the upper end surfaces of guide pins 302 and the upper surface of the substrate 220 are flush with each other.

Then, pair of guide pins 302 inserted into two through-holes 222 of substrate 220 positioned within the depth of field of the imaging device is imaged by imaging device 230. In other words, pair of guide pins 302 of the cut-and-clinch unit and two through-holes 222 of the substrate with pair of guide pins 302 inserted therein are simultaneously imaged by one imaging device 230 from above. Note that, as described above, imaging device 230 is a two-dimensional imaging device, and the depth of field of imaging device 230 is relatively narrow. Therefore, the upper ends of guide pins 302 inserted into through-holes 222 are positioned to be flush with the upper surface of substrate 220, so that an appropriate captured image of the upper ends of guide pins 302 and through-holes 222 is acquired.

Then, the amount of deviation between inserted guide pins 302 and through-holes 222 is calculated based on imaging data of two through-holes 222 with pair of guide pins 302 inserted therein. That is, for example, in a case where first guide pin 302a is inserted into first through-hole 222a and second guide pin 302b is inserted into second through-hole 222b as illustrated in FIG. 10, the amount of deviation ($\Delta X1$, $\Delta Y1$) in the XY-directions between the center of first guide pin 302a and the center of first through-hole 222a and the amount of deviation ($\Delta X2$, $\Delta Y2$) in the XY-directions between the center of second guide pin 302b and the center of second through-hole 222b are calculated. In addition, the angle of deviation ($\theta1$) formed by a straight line connecting the center of first guide pin 302a and the center of second guide pin 302b and a straight line connecting the center of first through-hole 222a and the center of second through-hole 222b is also calculated.

Subsequently, through-holes 222 into which two guide pins 302 are inserted are changed and other pair of through-holes 222 is imaged. For example, cut-and-clinch unit 300 is rotated by 90 degrees by the operation of rotation device 156 of unit moving device 36. Then, according to the above-described procedure, two guide pins 302 are inserted into two through-holes 222. At this time, as illustrated in FIG. 11,

13

first guide pin **302a** is inserted into first through-hole **222a**, and second guide pin **302b** is inserted into second through-hole **222c**. Then, the amount of deviation ($\Delta X3$, $\Delta Y3$) in the XY-directions between the center of first guide pin **302a** and the center of first through-hole **222a** and the amount of deviation ($\Delta X4$, $\Delta Y4$) in the XY-directions between the center of second guide pin **302b** and the center of second through-hole **222c** are calculated. In addition, the angle of deviation ($\theta 2$) formed by a straight line connecting the center of first guide pin **302a** and the center of second guide pin **302b** and a straight line connecting the center of first through-hole **222a** and the center of second through-hole **222c** is also calculated.

Furthermore, through-holes **222** into which two guide pins **302** are inserted are changed and other pair of through-holes **222** is imaged. For example, cut-and-clinch unit **300** is further rotated by 90 degrees by rotation device **156**. Then, according to the above-described procedure, two guide pins **302** are inserted into two through-holes **222**. Although not illustrated, first guide pin **302a** is inserted into first through-hole **222b**, and second guide pin **302b** is inserted into second through-hole **222a**. Then, the amount of deviation in the XY-directions between the center of first guide pin **302a** and the center of first through-hole **222b** and the amount of deviation in the XY-directions between the center of second guide pin **302b** and the center of second through-hole **222a** are calculated. In addition, an angle formed by a straight line connecting the center of first guide pin **302a** and the center of second guide pin **302b** and a straight line connecting the center of first through-hole **222a** and the center of second through-hole **222b** is also calculated.

Then, a correction value at the time of inserting pair of guide pins **302** into the pair of through-holes is acquired based on the calculated amount of deviation and angle of deviation. By using this correction value, it is possible to insert two guide pins **302a** and **302b** into the centers of two through-holes **222a** and **222b** as illustrated in FIG. 9, in a case where two guide pins **302a** and **302b** are to be inserted into two through-holes **222a** and **222b**, as illustrated in FIG. 9, for example. In other words, two guide pins **302** can be inserted into two through-holes **222** in a state where the amount of deviation and the angle of deviation are zero. As a result, it is possible to improve the accuracy of insertion at the time of insertion of guide pins **302** into through-holes **104** of circuit base material **12**.

Note that, as with acquisition of a correction value that is performed by using jig **200**, the imaging of pair of through-holes **222** with pair of guide pins **302** inserted therein, calculation of the amount of deviation and the angle of deviation based on imaging data acquired by the imaging of pair of through-holes **222**, acquisition of a correction value for insertion of the pair of guide pins into the pair of through-holes that is acquired based on the calculated amount of deviation and angle of deviation, or the like is performed at the time of inspection of the accuracy of component mouter **10** or at the time of maintenance, inspection, or the like of component mouter **10** after delivery to a user. As a result, it is possible to stabilize the quality of component mouter **10**, avoid problems, maintain the insertion accuracy by periodic maintenance, and the like.

Note that component mouter **10** is an example of a substrate working device. Circuit base material **12** is an example of a substrate. Component mounting device **24** is an example of an inserting device. Clamp device **52** is an example of a holding device. Through-hole **104** is an example of a through-hole. Lead component **106** is an example of a component. Lead **108** is an example of a

14

terminal. Jig **200** is an example of a jig. Component main body **202** is an example of a main body. Pin **204** is an example of a pin. Substrate **220** is an example of a substrate. Through-hole **222** is an example of a through-hole. Imaging device **230** is an example of an imaging device. Cut-and-clinch unit **300** is an example of an inserting device. Guide pin **302** is an example of a pin.

In addition, the present disclosure is not limited to the above-described embodiments, and can be implemented in various forms with various changes and improvements based on the knowledge of those skilled in the art. For example, in the above-described embodiment, pair of pins **204** and two through-holes **222** with pair of pins **204** inserted therein are simultaneously imaged by one imaging device **230** from a lower side of a vertical axis. In addition, pair of guide pins **302** and two through-holes **222** with pair of guide pins **302** inserted therein are simultaneously imaged by one imaging device **230** from an upper side of the vertical axis. Meanwhile, in a case where pair of pins **204** and two through-holes **222** are simultaneously imaged by imaging device **230**, the relative positions of the pair of pins and the pair of through-holes and the angle of alignment, that is, the amount of positional deviation and the angle of deviation can be calculated. In addition, in a case where pair of guide pins **302** and two through-holes **222** are simultaneously imaged by imaging device **230**, the relative positions of the pair of guide pins and the two through-holes and the angle of alignment, that is, the amount of positional deviation and the angle of deviation can be calculated. That is, an image captured by imaging device **230** simultaneously imaging pair of pins **204** and two through-holes **222** with the pair of pins not inserted therein may be acquired, or an image captured by imaging device **230** simultaneously imaging pair of guide pins **302** and two through-holes **222** with the pair of guide pins not inserted therein may be acquired. However, both of pair of pins **204** or the distal ends of guide pins **302** and two through-holes **222** need to be in a state of being able to be simultaneously imaged in a viewpoint from one imaging device **230** that is fixedly disposed and need to be not concealed.

In addition, in the above embodiment, component mounting device **24** for inserting pins **204** into through-holes **222** and imaging device **230** for imaging through-holes **222** with the pins inserted therein are disposed at opposite positions with respect to substrate **220**. In addition, cut-and-clinch unit **300** for inserting guide pins **302** into through-holes **222** and imaging device **230** for imaging through-holes **222** with the guide pins inserted therein are disposed at opposite positions with respect to substrate **220**. Meanwhile, both of component mounting device **24** and imaging device **230** may be disposed above substrate **220** and both of cut-and-clinch unit **300** and imaging device **230** may be disposed below substrate **220**. That is, in a case where pins **204** are inserted from a position above the substrate **220**, through-holes **222** with pins **204** inserted therein may be imaged above substrate **220** at an angle offset from the vertical axis. In addition, in a case where guide pins **302** are inserted from a position below the substrate **220**, through-holes **222** with guide pins **302** inserted therein may be imaged below substrate **220** at an angle offset from the vertical axis.

In addition, although pins **204** of jig **200** extend linearly and vertically from component main body **202** in the above embodiment, pins **204** may be bent or extend from the component main body in a state of being inclined as long as pins **204** are standardized pins, that is, as long as the positions of the pins are known in advance. Although component holding tool **77** grips component main body **202**

of jig **200** in the above embodiment, component holding tool **77** may grip a pin. In addition, a jig without a component main body may be adopted. For example, a rod-shaped member may be bent into a U-shape, both ends of the member may be used as substitutes for a pair of pins, and the member may be gripped by a component holding tool and used as a jig. A correction value at the time of acquisition with the component holding tool gripping the pins as described above is used as a correction value used when the component holding tool grips pins of a component and inserts the pins into through-holes of a circuit substrate. In other words, a jig including a pin that is not standardized may be adopted. However, in a case where a jig including a pin that is not standardized is used, it is necessary to image the pin with an imaging device to recognize the position of the pin. In other words, it is possible to use commercially available lead component **106** as jig **200** without using a standardized jig.

In addition, although through-holes **222** are imaged with substantial pins **204** inserted into through-holes **222** in the above embodiment, insubstantial pins (for example, illustrations of pins printed on the jig), holes processed in the jig, or the like may be used instead as substitutes for the pins of the jig. In such a case, for example, illustrations or the like of a linear shape or a rod shape resembling a pin, a lead, or the like may be printed on the component main body of the jig, a pair of holes may be drilled in the component main body of the jig, and the illustrations or the holes and through-holes **222** may be simultaneously imaged for acquisition of a correction value. That is, whether the pins are substantial does not matter as long as shapes, from which the shapes of the pins can be figured out (in other words, shapes of the pins), and images of through-holes **222** are simultaneously captured by imaging device **230**. As described above, jigs of various shapes can be adopted. In addition, various types of component holding tools such as a suction nozzle picking up a main body may be adopted depending on the shape of the jig.

Although the amount of deviation between the center of pin **204** or the guide pin and the center of through-hole **222** is calculated in the above embodiment, various values based on imaging data may be calculated instead of the amount of deviation between the centers as long as the relative positions of pin **204** or the guide pin and through-hole **222** can be figured out. For example, the distance between an inner wall of through-hole **222** and an external line of the lower end surface of pin **204** may be calculated. Note that the same applies to the relative positions of guide pin **302** and through-hole **222**.

Although one piece of two-dimensional image data captured by one two-dimensional imaging device is used in the above embodiment, imaging data captured by another type of imaging device (for example, a stereo camera, a video camera, and a three-dimensional camera) or multiple imaging devices may be used. In addition, imaging data captured by a stereo camera, a video camera, and a three-dimensional camera may be intentionally converted into two-dimensional imaging data and the two-dimensional imaging data may be used.

In addition although an operation of mounting a radial component as lead component **106** has been described in the above embodiment, the present disclosure can also be applied to an axial component. In addition, the present disclosure is not limited to a lead component, and can be applied to a component including a terminal to be inserted into through-hole **222**.

REFERENCE SIGNS LIST

10: component mounter (substrate working machine), **12**: circuit base material (substrate), **24**: component mounting device (inserting device), **52**: clamp device (holding device), **104**: through-hole, **106**: lead component (component), **108**: lead (terminal), **200**: jig, **202**: component main body (main body), **204**: pin, **220**: substrate, **222**: through-hole, **230**: imaging device, **300**: cut-and-clinch unit (inserting device), **302**: guide pin (pin)

The invention claimed is:

1. A substrate working machine comprising:

a base material conveyance and holding device configured to convey and hold a substrate in which multiple through-holes are formed, the base material conveyance and holding device including a conveyance device and a clamp device;

an component mounting device configured to insert multiple terminals of a component into the multiple through-holes of the substrate held by the base material conveyance and holding device, the component mounting device including a work head and a work head moving device;

a jig configured to be held by the component mounting device, the jig including a main body and a pair of pins extending from the main body, a shape of the pair of pins being standardized; and

a camera configured to simultaneously image the pair of pins and a pair of through-holes of the multiple through-holes; and

a controller including a processor configured to control the component mounting device so that XY-coordinates of the pair of pins of the jig held by the component mounting device coincide with XY-coordinates of first and second through-holes of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and control the camera to acquire a first image data, control the component mounting device so that XY-coordinates of the pair of pins of the jig coincide with XY-coordinates of one of the first or second through-holes and a third through-hole of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and control the camera to acquire a second image data, calculate positions of the first, second, and third through-holes and the shape of the pair of pins based on the acquired first and second image data captured by the camera,

calculate translational and angular deviation amounts between centers of the first and second through-holes and centers of the pair of pins, and centers of the one of the first or second through-holes and the third through-hole and centers of the pair of pins, calculate a correction value for inserting multiple terminals of a component into the multiple through-holes of the substrate by the component supply device based on the calculated deviation amounts, and control the component supply device to insert multiple terminals of a component into the multiple through-holes of the substrate based on the correction value.

2. The substrate working machine according to claim 1, wherein the image data is two-dimensional image data.

17

3. A method comprising:
 providing the substrate working machine according to claim 1;
 providing a substrate;
 controlling the component mounting device so that XY-coordinates of the pair of pins of the jig held by the component mounting device coincide with XY-coordinates of first and second through-holes of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an up-down direction, and controlling the camera to acquire a first image data;
 controlling the component mounting device so that XY-coordinates of the pair of pins of the jig coincide with XY-coordinates of one of the first or second through-holes and a third through-hole of the multiple through-holes of the substrate, and end surfaces of the pair of pins are flush with a lower surface of the substrate in an

18

up-down direction, and controlling the camera to acquire a second image data;
 calculating positions of the first, second, and third through-holes and the shape of the pair of pins based on the acquired first and second image data captured by the camera;
 calculate translational and angular deviation amounts between centers of the first and second through-holes and centers of the pair of pins, and centers of the one of the first or second through-holes and the third through-hole and centers of the pair of pins calculate a correction value for inserting multiple terminals of a component into the multiple through-holes of the substrate by the component supply device based on the calculated deviation amounts, and control the component supply device to insert multiple terminals of a component into the multiple through-holes of the substrate based on the correction value.

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